A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 12

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 4 points below. **1 Vehicles:** high energy density (200 Wh kg^{-1}) minimises battery mass so range is greater – many cycles and high efficiency **1 Stationary storage:** mass is unimportant so the low energy density of lead-acid does not matter – lowest cost per kWh – well-established, recyclable **1 Lead-acid drawbacks:** fewer cycles and lower efficiency – lithium fire risk needs management **1** Nickel-cadmium largely replaced because of cadmium toxicity and cost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4

Do not write outside the box

Question	Marking guidance	Marks
01.2	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Recharging speed **1** Self-discharge during non-use **1** Availability of raw materials (lithium, cobalt) **1** Recyclability / disposal **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Parked electric vehicles plugged into the grid form a very large distributed battery **1** At peaks a small proportion of their stored energy is fed back to the grid – surplus grid energy recharges them - avoiding costly, low-efficiency open-cycle gas turbines **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Surplus electricity electrolyses water to hydrogen, which can be converted to methane **1** Methane is fed into the existing natural gas pipe network and storage, so no new distribution system is needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 223-224; AQA spec 3.3.4.1.	
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** 1 nitrogen fixation – 2 nitrification **1** 3 denitrification – 4 ammonification **1** 5 ionisation – 6 leaching (3 marks for all six, 2 for four or five, 1 for two or three) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **nitrification** → ammonium oxidised to nitrite and then nitrate • **denitrification** → nitrate converted back to nitrogen gas	3

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** Drainage makes the soil aerobic **1** Aerobic nitrifying bacteria (and nitrogen fixers) increase so more ammonium is converted to nitrate **1** Anaerobic denitrifying bacteria decline so less nitrate is lost as nitrogen gas **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **waterlogged** → soil pores filled with water so there is no air • **nitrifying** → ammonium oxidised to nitrite and then nitrate • **denitrifying** → nitrate converted back to nitrogen gas	3
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Nitrogen gas is very unreactive so most organisms cannot use it directly **1** Only nitrogen fixation (bacteria, lightning, Haber process) converts it to usable ammonium / nitrate – the cycle recycles a small usable amount **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds	2
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Rhizobium bacteria in their root nodules fix atmospheric nitrogen into ammonium / proteins **1** When the plants die / are ploughed in, decomposition and nitrification release nitrates for the following crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrification** → ammonium oxidised to nitrite and then nitrate	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 152-154; AQA spec 3.2.4.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** There is an environmental gradient (distance from the hedge / changing conditions) so samples must be taken at regular intervals along it to detect the change **1** Random sampling might leave gaps / miss the pattern – systematic sampling avoids bias while covering the gradient **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **systematic sampling** → sampling at regular intervals • **transect** → a line along which samples are taken to show change with distance • **random** → positions chosen using random numbers so the choice is not biased **Do not credit** • 'pick places at random by eye' - positions have to be generated with random numbers to remove bias	2
03.2	**1 mark each** 1 mark for each of the 4 points below. **1** Sample at close intervals (eg every 1 m) first and see at what spacing the variations are still detected without unnecessary data – wider intervals (eg 4 m) may miss peaks and troughs **1** Take increasing numbers of samples at one site and calculate the running mean – the number needed is where the mean stops fluctuating / stabilises **1** Calculate standard deviation to assess scatter: more variable data need more samples **1** Test the method (extraction technique, timing) for reliability before the main study **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **standard deviation** → a measure of the spread of the data about the mean • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	4

Do not write outside the box

Question	Marking guidance	Marks
03.3	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2** Remove surface vegetation – pour an irritant solution (dilute mustard / detergent in water) onto the area until saturated (and around it), keep adding as it infiltrates **up to 2** Collect worms that come to the surface for a standardised time – count / identify / weigh **up to 2 Limitations:** solution may not percolate impermeable soils or reach deep worms – worms may move sideways or downwards – some may die or not respond – slow-moving worms missed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393; AQA spec 3.7.1.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended **1** Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges **1** The particles clump together (flocs) and settle in the clarifier tank **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources **1** Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed – improves taste **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) **1** Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**1 mark each** 1 mark for each of the 3 points below. **1 Ion exchange:** toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) **1** Reverse osmosis / distillation (desalination) - expensive, energy-intensive **1 Activated carbon:** organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
05.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1
05.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	1 mark for the point below. **1** Temperature inversion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Temperature inversion** → a layer of warm air above cooler air that traps pollutants	1
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** On night A, temperature falls with altitude – warm effluent gases are less dense / more buoyant than the surrounding air so they rise **1** As they rise they cool but the surrounding air is also cooler so they stay buoyant and continue to rise, dispersing and diluting **1** On night B the air near the ground is colder / denser than the air above (inversion layer at about 400 m) **1** Pollutants cool as they rise and become denser than the warmer air above so they are trapped below the inversion layer and become concentrated **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
06.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Cloudless sky allows infrared energy to be radiated from the ground so the ground cools rapidly **1** The cold ground cools the air in contact with it, producing a cold dense layer near the ground **1** Low wind velocity means the air layers of different temperatures do not mix **1** Valley allows cold dense air to collect / drain into the low ground **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	3
06.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Smoke smogs (eg London 1952) **1** Photochemical smogs (eg Los Angeles, Mexico City) **1** Build-up of NOx / carbon monoxide / PM10 in cities **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **PM10** → fine solid particles / smoke and dust **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 241-242; AQA spec 3.4.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 4 points below. **1 Gasification:** coal (including deposits too deep to mine) is burnt underground under controlled conditions to produce fuel gases (hydrogen, carbon monoxide, methane) **1** Allows deep coal to be exploited and gas can be piped / used in gas turbines or for hydrogen with CCS **1 Liquefaction:** coal is converted to liquid hydrocarbons directly with solvents or via gasification and chemical conversion **1** Provides liquid fuels for vehicles / aircraft that solid coal cannot supply **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CCS** → capturing carbon dioxide from flue gases and storing it underground	4
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** A solid, ice-like crystalline structure in which methane is trapped within water ice **1** Found at low temperatures (polar regions / permafrost) or high pressure in ocean sediments around continental margins **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.3	**1 mark each** 1 mark for each of the 4 points below. **1 Water heating:** hot water pumped into sediments melts the hydrate releasing methane **1 Depressurisation:** drilling reduces pressure so methane dissociates from the crystals **1 CO2 injection:** CO2 bonds more strongly to the ice and displaces methane (could double as CCS) **1 Risks:** methane not collected escapes to the atmosphere - powerful greenhouse gas / positive feedback – seabed ecology poorly understood – sediment instability / landslides **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **CCS** → capturing carbon dioxide from flue gases and storing it underground • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187, 190; AQA spec 3.3.4.1.	
08.1	1 mark for the point below. **1** As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 2400 / 40 = 60 times **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
08.3	**1 mark each** 1 mark for each of the 3 points below. **1 Reserves:** the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) **1 Resource:** all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits **1** The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	3
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Lowering the cut-off ore grade includes deposits of lower purity in the reserves **1** Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.5	**1 mark each** 1 mark for each of the 3 points below. **1** The total quantity (stock / resource base) is far greater than could ever be used **1** But most is at purities too low or in locations too inaccessible for current economic exploitation **1** The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable – energy needs rise rapidly as ore grade falls **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** pH 5.6 **1** Atmospheric carbon dioxide dissolves in rainwater to form (weak) carbonic acid **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
09.2	**1 mark each** 1 mark for each of the 4 points below. **1** Limestone bedrock (calcium carbonate) is alkaline / basic and neutralises the acids as water percolates through it, so the lake pH stays high (7.2) **1** Granite does not neutralise acids so lake water in A is acidic (4.6) **1** Low pH mobilises / dissolves aluminium ions from the soil and rock (0.42 mg dm^{-3} in A) which are toxic to fish (damage gills) **1** Acids also damage fish eggs / gills (exposed living cells) and dissolve the exoskeletons of invertebrate food – lichens and other food organisms lost **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
09.3	**1 mark each** 1 mark for each of the 2 points below. **1 Corrodes metals:** railway lines, railings, pipes, pylons, overhead power lines **1** Dissolves / weakens limestone and marble buildings and statues (eg York Minster) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Sulfur dioxide is oxidised (by ozone / other oxidants) to sulfur trioxide **1** Sulfur trioxide dissolves in water droplets to form sulfuric acid (SO2 alone gives the weaker sulfurous acid) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 250-252; AQA spec 3.4.3.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Properties of renewables:** intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics • **Wind:** large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments – intermittent, needs storage and grid – land-use conflicts • **Solar:** PV costs falling rapidly, roof space, but low winter insolation in the UK – CSP unsuitable – heat pumps and photothermal for heating • **HEP:** most large sites used – low-head, helical and micro-hydro extend use – pumped storage vital • **Tidal:** large range (Severn, Mersey), predictable – barrages costly with high impact, lagoons and in-stream turbines lower impact • **Wave:** long Atlantic fetch, but technology immature • **Biofuels:** wastes, anaerobic digestion, energy crops vs food / habitat • **Geothermal:** limited to low-temperature schemes / heat pumps • **Systems issues:** storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport • **Judgement:** renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder – transition needs investment and lifestyle change **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any accurate, relevant material that is not in the indicative content. • **intermittency** → only generates when the wind blows or the Sun shines / supply is not continuous • **insolation** → incoming solar radiation / energy arriving from the Sun • **HEP** → electricity generated from falling water **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	
	Model answer - Level 3 (8 marks)	
	What renewables can contribute depends on their properties - intermittency, predictability, energy density, storage, applicability, location, impacts, technological maturity and cost - set against the UK's resources. Wind is the largest: shallow seas give a huge offshore resource, costs have fallen and direct-drive and vertical-axis designs keep improving, but it is intermittent, needs storage and grid connections and provokes land-use conflicts onshore. Solar PV costs are falling rapidly and roof space is plentiful, but winter insolation is low and concentrating solar is unsuitable; photothermal panels and heat pumps are better for heating. Most large HEP sites are used, though low-head, helical and micro-hydro extend it and pumped storage is vital for balancing. The Severn and Mersey have large, predictable tidal ranges, but barrages are costly and damaging while lagoons and in-stream turbines have lower impact; the long Atlantic fetch gives wave energy that the technology cannot yet harness reliably. Biofuels from wastes and anaerobic digestion are useful, but energy crops compete with food and habitat, and geothermal is limited to low-temperature schemes.	
	The system matters as much as the sources: storage in pumped HEP, batteries, hydrogen and power-to-gas, vehicle-to-grid, grid management, a diverse mix that smooths supply, and conservation to reduce demand. With these, renewables could supply most of the UK's electricity. Heat and transport are harder, because they need high temperatures or liquid fuels, so electrification and hydrogen would be required. The potential is large, but realising it needs sustained investment and changes in how energy is used.	
	Weaker answer - Level 1 (3 marks)	
	The UK could use lots of renewable energy. It is windy so wind turbines could be built, especially offshore. Solar panels can be put on roofs but it is not very sunny in winter. Tidal power could be used in the Severn estuary because it has big tides. Hydroelectric power can be used in Scotland. Renewables are intermittent so batteries would be needed. Renewables could provide a lot of the UK's energy in the future.	
	What makes the difference	
	• Level 1 mentions the main resources with one fact each; Level 3 assesses each against its properties, constraints and the state of the technology.	
	• Level 3 treats storage, grid and demand as a system and distinguishes electricity from heat and transport - essential for judging 'future energy supplies'.	

Do not write outside the box

Question	Marking guidance	Marks
	• The conclusion in Level 3 is qualified (most electricity; heat and transport harder; needs investment); Level 1's is unqualified.	
10.2	1 mark for the point below. **1** Eight times (energy proportional to velocity cubed) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	
11.1	**Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Why non-renewable:** formed too slowly – geological concentration processes - hydrothermal, metamorphic, Proterozoic sediments, alluvial, evaporites, secondary enrichment, biological sediments • Understanding processes guides exploration (batholith margins, river gravels, ancient seas) and remote sensing / geophysical techniques • Reserves, resource, stock – Lasky's principle – cut-off ore grade and its drivers • **Extraction technologies:** mechanisation, bioleaching, phytomining, polymer and bacterial adsorption, seabed nodules • **Reducing demand:** recycling and its difficulties, Cradle to Cradle, design for disassembly, linear vs circular material flows, dispersal of metals in use • **Environmental limits:** impacts of mining and their control – energy costs rising with lower grades – conflicts	25

Do not write outside the box

Question	Marking guidance	Marks
	• **Judgement:** technology can extend supplies substantially but only within energy and environmental constraints – circular economy essential *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 130-147, 373-379; AQA spec 3.2.3, 3.6.3-3.6.4.	
11.2	**Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Properties:** toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility – DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • **Reducing impact:** restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • **Alternatives:** cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • **Trade-offs:** yield, cost, human health (organophosphates), resistance evolution, pollinator services – risk:benefit – sustainability and the circular economy	25

Do not write outside the box

Question	Marking guidance	Marks
	• **Judgement:** no pesticide is perfectly specific – combination of property-based selection and integrated control is most sustainable *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	

Do not write outside the box